

2012 Revised Higher Physics: scan the code and answer these.



Q17

Q18

Q20

Section A · electricity

Show your working, not just the letter.

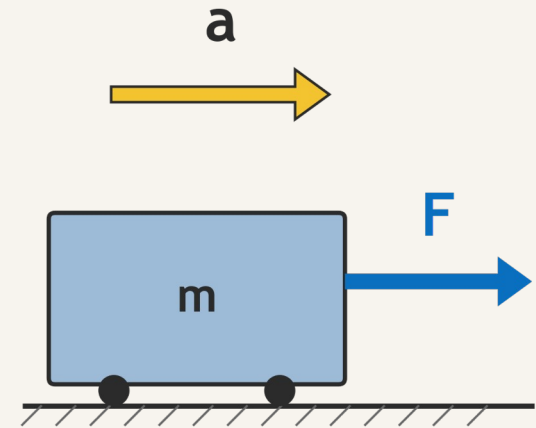


Scan for the paper
Marking scheme

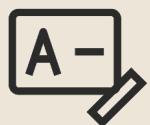
Forces

Why does anything ever speed up?

Higher Physics



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.

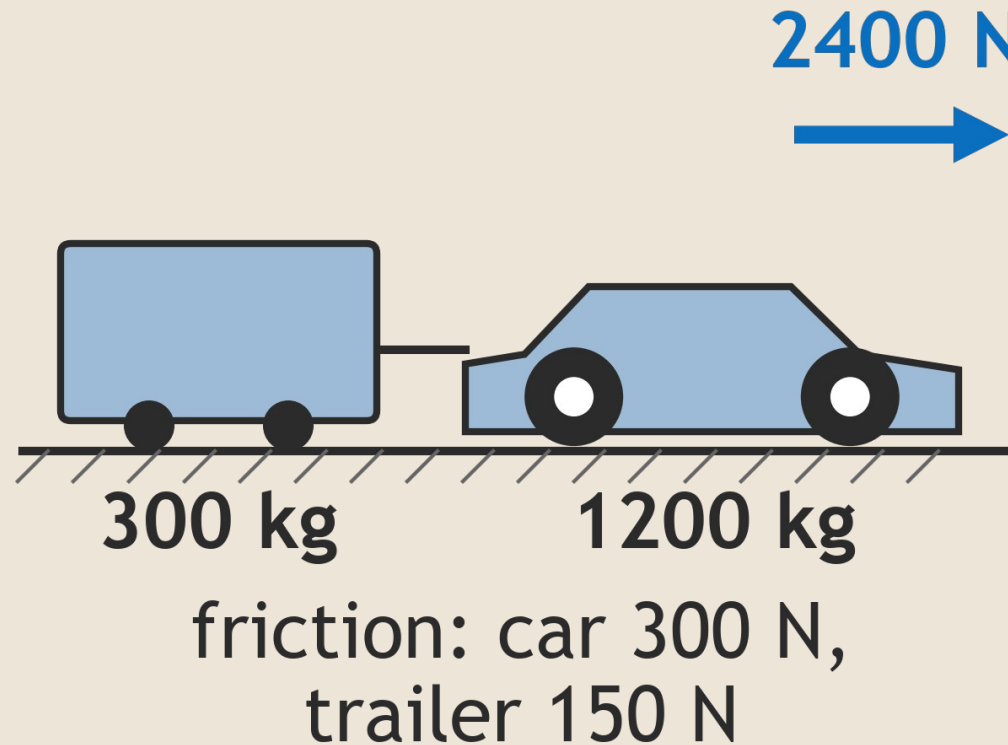


Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

A car tows a trailer. Is the pull on the trailer as big as the car's engine force?



PART 1

Balanced and unbalanced forces

Motion only changes when the forces stop balancing.

By the end of this part...



- 1 I can add forces as vectors to find the resultant force
- 2 I can use Newton's first and second laws to explain motion
- 3 I can use $F = ma$ with the resultant of several forces

Recap from N5: Forces

Balanced

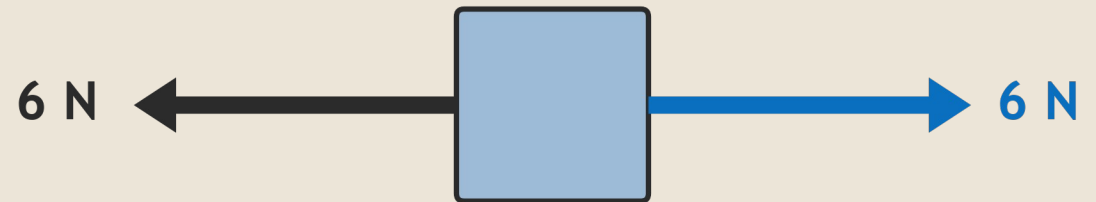
at rest or constant velocity

Unbalanced

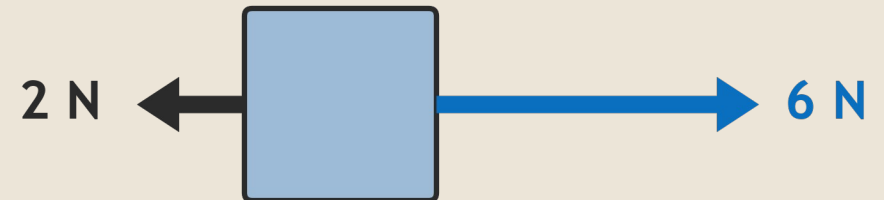
accelerates: $F = ma$

Free body diagram

only the forces ON the object



balanced: resultant = 0



unbalanced: resultant = 4 N →

About Isaac Newton

Born

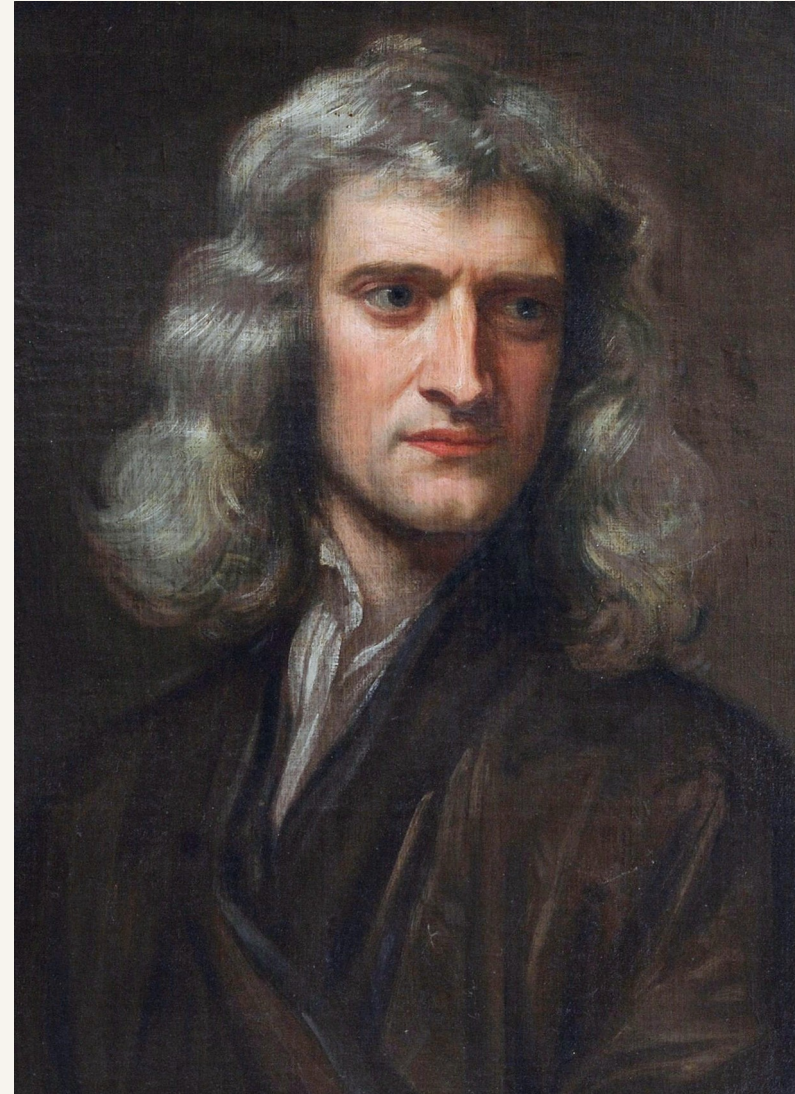
1643, Woolsthorpe,
Lincolnshire

Died

1727, London

Plague years

worked at home, 1665-66



Newton: Discoveries and Theories

Motion

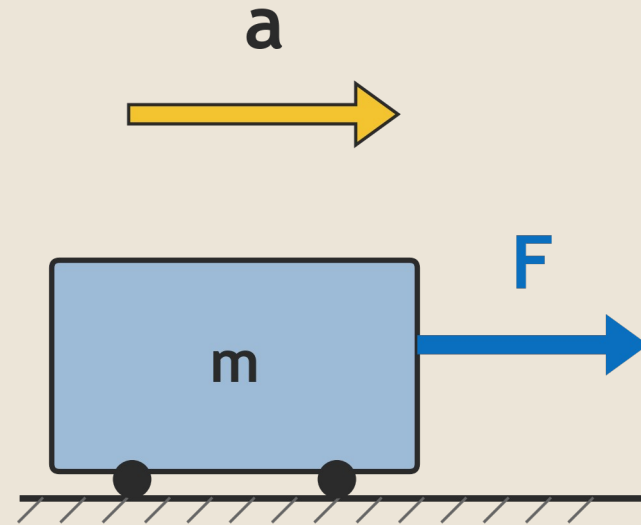
three laws of motion

Gravity

universal gravitation

Light

white light is a mix of colours



Recap: $F = ma$

$$F = ma$$

F

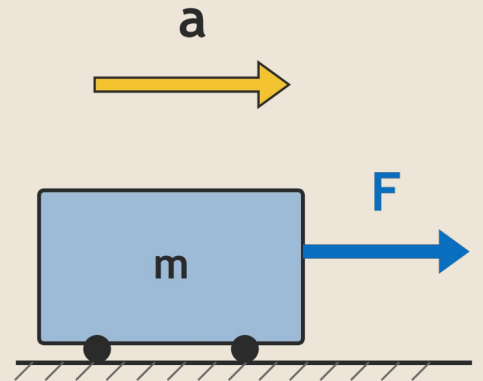
unbalanced
force (N)

m

mass (kg)

a

acceleration
(m s^{-2})



Recap: $W = mg$

$$W = mg$$

W

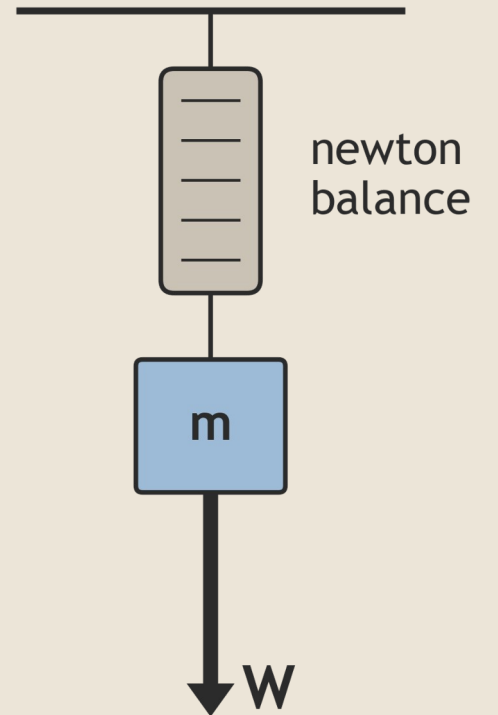
weight (N)

m

mass (kg)

g

gravitational
field strength
(N kg⁻¹)



Worked Example: Forces at Right Angles

Ropes pull a 50 kg crate across smooth ice: 30 N east and 40 N north. Calculate its acceleration.

4

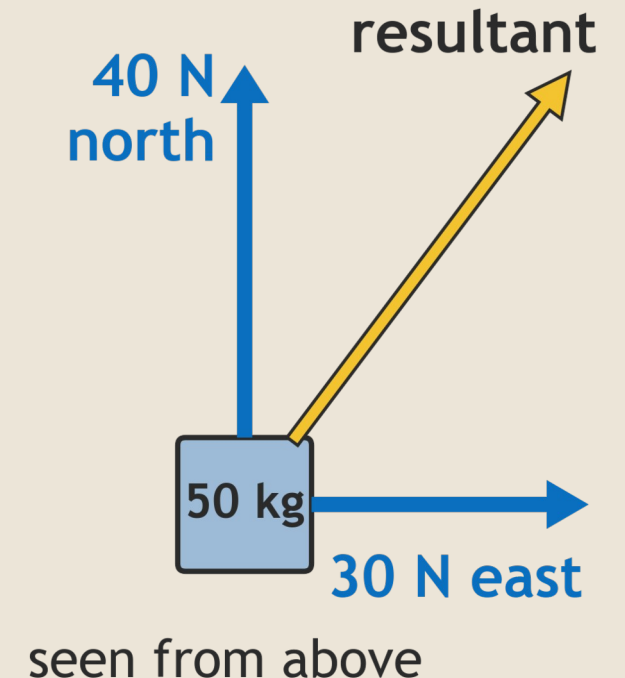
$$F = ma \quad (1)$$

$$(\text{resultant}^2 = 30^2 + 40^2, \text{ so } 50 \text{ N})$$

$$50 = 50 \times a \quad (1)$$

$$a = 1.0 \text{ m s}^{-2} \quad (1)$$

$$\tan \theta = 40 \div 30, \text{ so } 53^\circ \text{ north of east} \quad (1)$$



Worked Example: A Falling Ball

A 0.50 kg ball falls. Air resistance is 1.9 N. Calculate its acceleration.

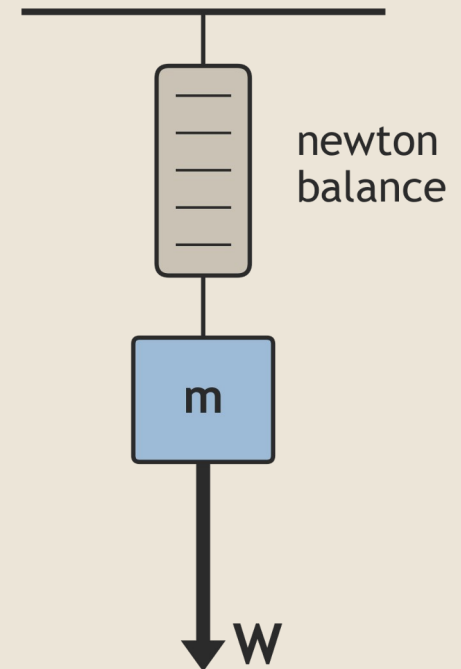
3

$$F = ma \quad (1)$$

$$(W = mg = 0.50 \times 9.8 = 4.9 \text{ N})$$

$$4.9 - 1.9 = 0.50 \times a \quad (1)$$

$$a = 6.0 \text{ m s}^{-2} \quad (1)$$



Your Turn: Forces



1

6.0 N east and 8.0 N north act on a 2.0 kg mass. Calculate its acceleration.

5.0 m s⁻² — so the resultant is 10 N

2

A 0.50 kg ball falls. Air resistance is 2.9 N. Calculate its acceleration.

4.0 m s⁻²

3

A 1200 kg car accelerates at 1.5 m s⁻². Friction is 600 N. Calculate the engine force.

2400 N — so ma + friction

2025 Paper 1 Q3. A force of 10.0 N pulls a trolley of mass 2.0 kg along a horizontal surface. The total frictional force on the trolley is 6.0 N. The acceleration is



A 2.0 m s⁻²

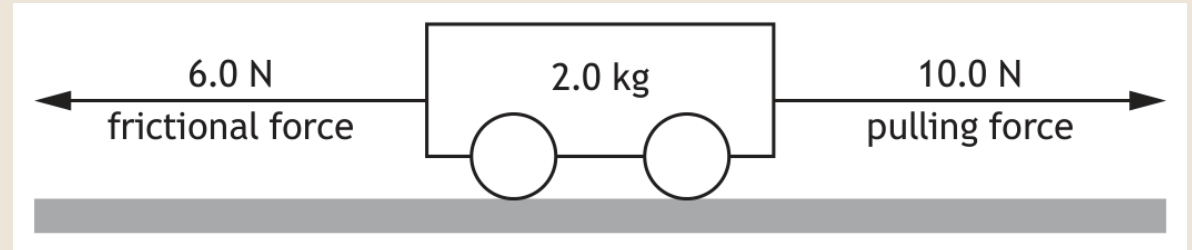
B 3.0 m s⁻²

C 5.0 m s⁻²

D 8.0 m s⁻²

E 10.0 m s⁻²

Answer A: $F = 10.0 - 6.0 = 4.0 \text{ N}$ — so $a = 4.0 \div 2.0$



Spot the mistake. A 3.0 kg trolley is pulled by 12 N.
Friction is 3.0 N. What is wrong with this answer?

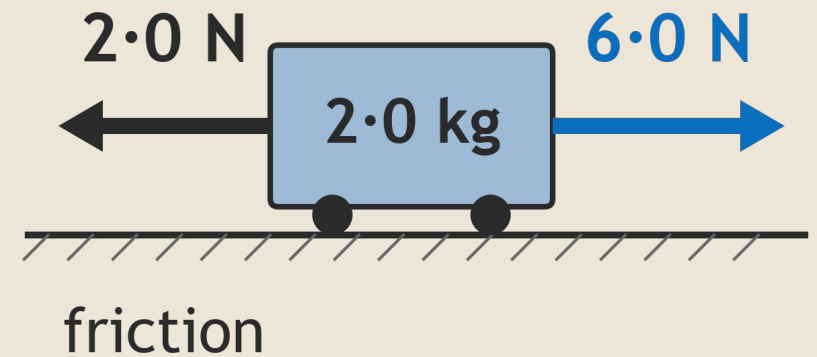


$$a = F \div m$$

$$a = 12 \div 3.0$$

$$a = 4.0 \text{ m s}^{-2}$$

**F is the unbalanced force — so $F = 9.0$
N and $a = 3.0 \text{ m s}^{-2}$**



Plenary: Forces and $F = ma$



1

5.0 N and 12 N act at right angles. Calculate the resultant.

13 N

2

Which force goes in $F = ma$?

The resultant force

3

A car moves at constant velocity. What is the resultant force?

Zero – so balanced

Look back — by the end of this part...



- 1 I can add forces as vectors to find the resultant force
- 2 I can use Newton's first and second laws to explain motion
- 3 I can use $F = ma$ with the resultant of several forces

2015 Higher Physics: scan the code and answer these.



Q8

Q10

Q16

Section A · particles, a.c., uncertainty

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 2

Newton's laws in action

Two blocks, one string: which mass do you use?

By the end of this part...



- 1 I can draw free body diagrams for connected objects
- 2 I can find the tension between connected objects, with friction

Connected Objects

Whole system

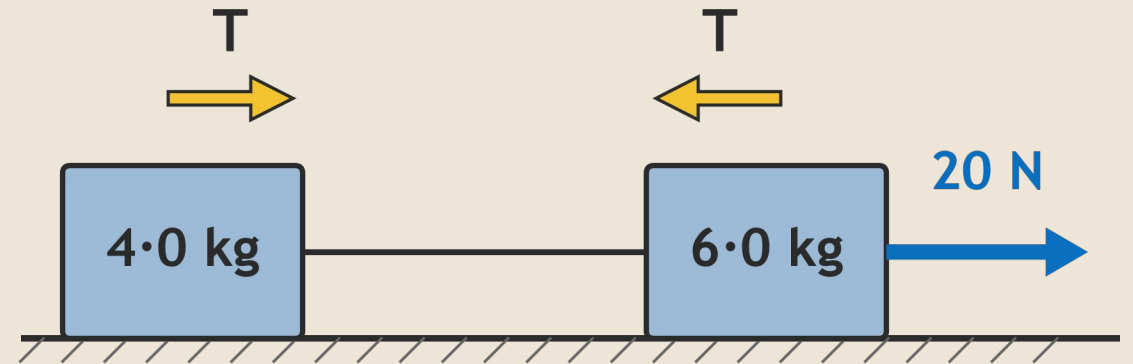
$$F = (m_1 + m_2) a$$

Tension T

the pull of the string on each block

One block

use only that block's mass



Worked Example: Two Trolleys

10 N pulls the 2.0 kg trolley. No friction. Calculate the tension in the string.

4

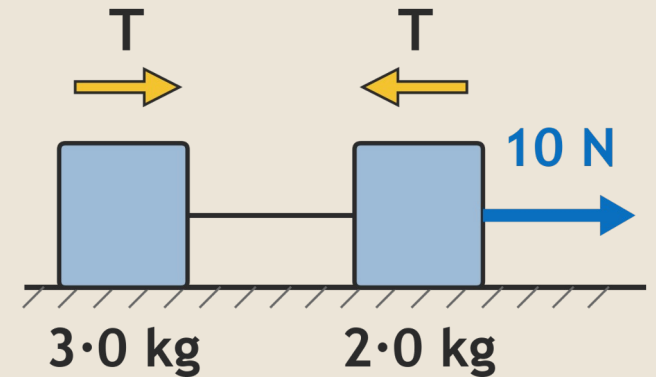
$$F = ma \quad (1)$$

$$10 = (3.0 + 2.0) \times a \quad (1)$$

$$a = 2.0 \text{ m s}^{-2}$$

$$T = 3.0 \times 2.0 \quad (1)$$

$$T = 6.0 \text{ N} \quad (1)$$



Worked Example: Tension with Friction

A car tows a trailer with the forces shown. Calculate the tension in the tow bar.

4

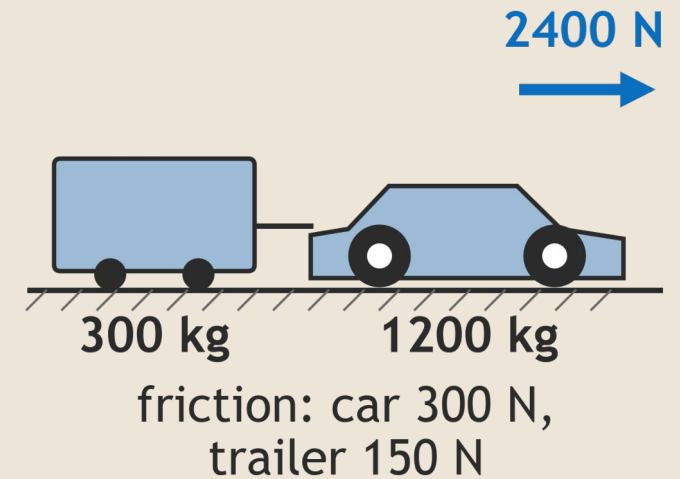
$$F = ma \quad (1)$$

$$2400 - 450 = 1500 \times a \quad (1)$$

$$(a = 1.3 \text{ m s}^{-2})$$

$$T - 150 = 300 \times 1.3 \quad (1)$$

$$T = 540 \text{ N} \quad (1)$$



Your Turn: Connected Objects



1

A 1000 kg tractor pulls a 500 kg trailer with a 900 N force. No friction. Calculate the tension.

300 N

2

The same, but 100 N friction acts on the trailer only.

370 N – so add the friction

3

10 N pulls a 3.0 kg block, which pulls a 2.0 kg block. No friction. Calculate T.

4.0 N – so the 2.0 kg block's mass

2015 Section 1 Q5. A 900 kg car pulls a 400 kg caravan with an acceleration of 2.0 m s^{-2} . Friction on the caravan is negligible. The tension in the coupling is



A 400 N

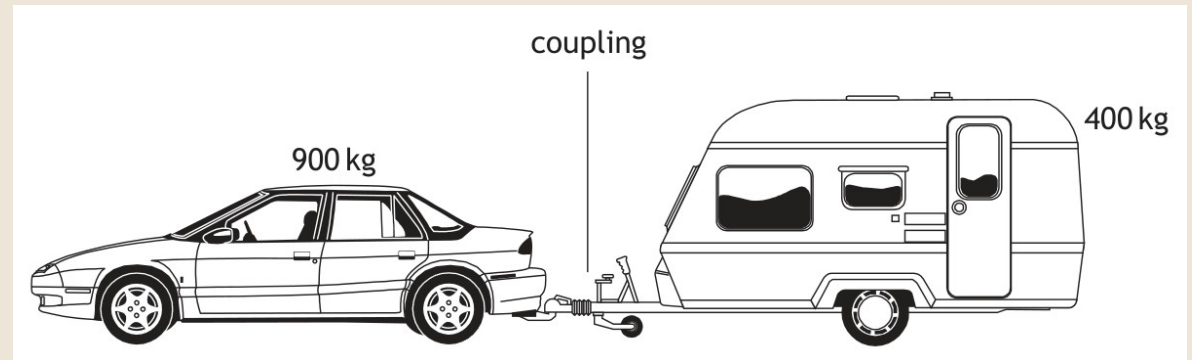
B 500 N

C 800 N

D 1800 N

E 2600 N

Answer C: $T = 400 \times 2.0$ — so the caravan's mass only



Spot the mistake. A 1200 kg car tows a 600 kg trailer at 0.50 m s^{-2} . No friction on the trailer. What is wrong with this answer?



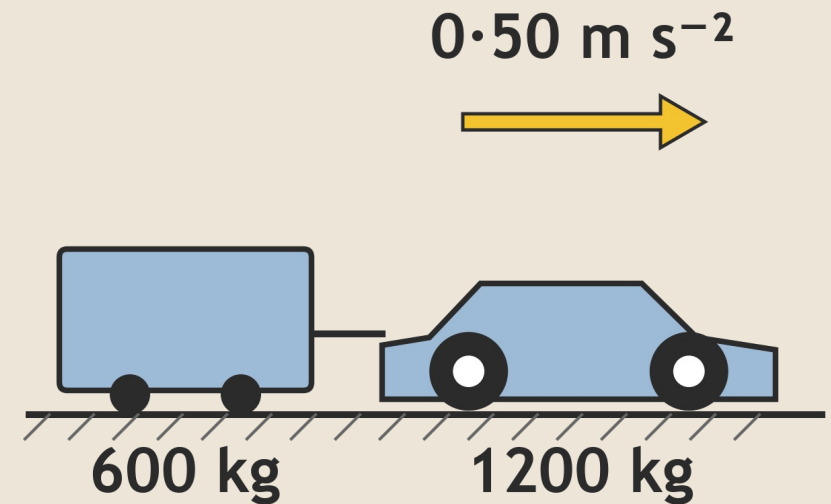
$$T = ma$$

$$T = (1200 + 600) \times 0.50$$

$$T = 900 \text{ N}$$

The tow bar pulls only the trailer – so

$$T = 600 \times 0.50 = 300 \text{ N}$$



Plenary: Newton's Laws in Action



- 1 2.0 kg and 3.0 kg blocks are pulled by 10 N. No friction. Calculate a.
- 2 For the tension in a string, which mass do you use?
- 3 Friction acts on a towed trailer. Does T go up or down?

$$2.0 \text{ m s}^{-2}$$

Only the block the string pulls

Up — so $T = ma + \text{friction}$

Look back — by the end of this part...



- 1 I can draw free body diagrams for connected objects
- 2 I can find the tension between connected objects, with friction

2014 Higher Physics: scan the code and answer these.



Q2

Q3

Q9

Section A · motion, circuits

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 3

Lifts and apparent weight

Same pupil, same mass. The scales show three different readings.

By the end of this part...



- 1 I can draw the forces on a person standing on scales in a lift
- 2 I can use $F = ma$ to find a scales reading or a lift's acceleration
- 3 I can describe how to measure a lift's acceleration with newton scales

Why do you feel heavier as a lift starts to go up?



A lift



A drop tower ride

What the Scales Read

Scales

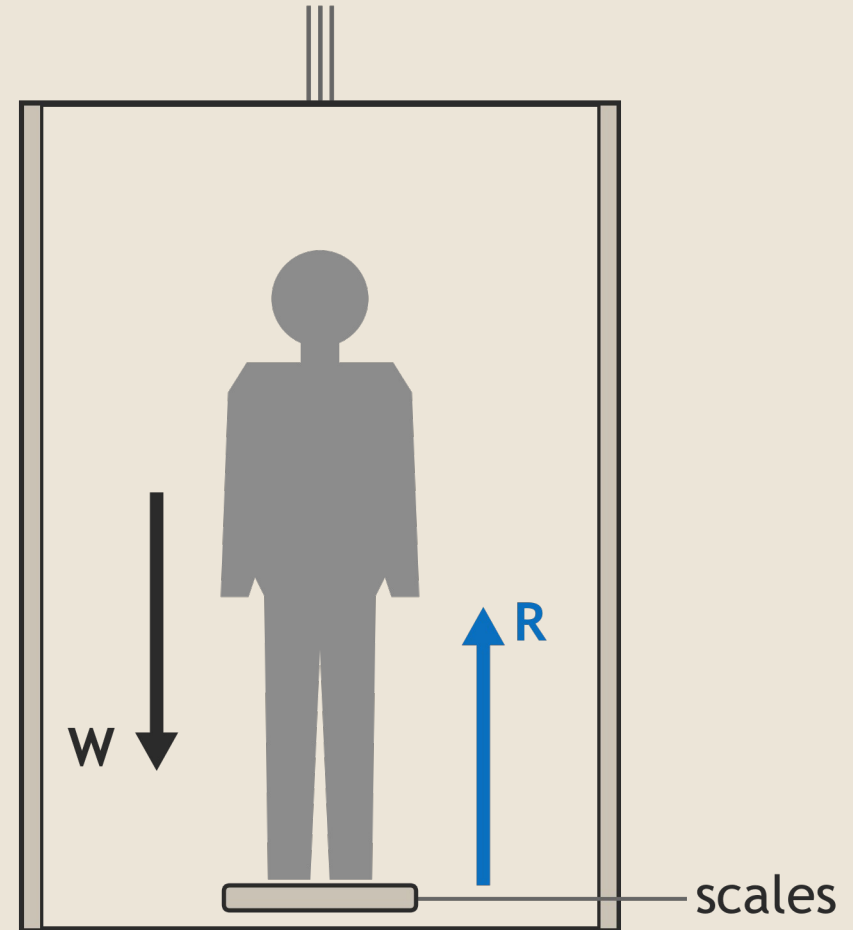
read R , the push up on you

Weight

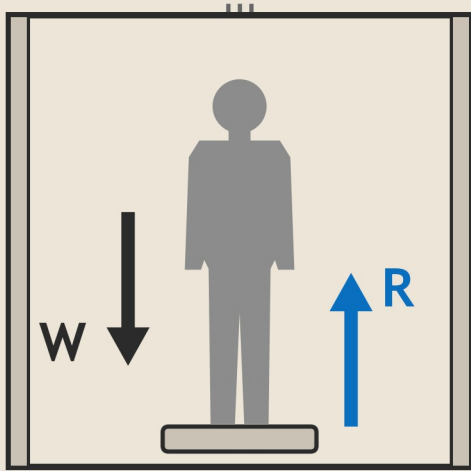
$W = mg$, down, never changes

Also called

a newton balance



A Lift Going Up



$$v = 0$$

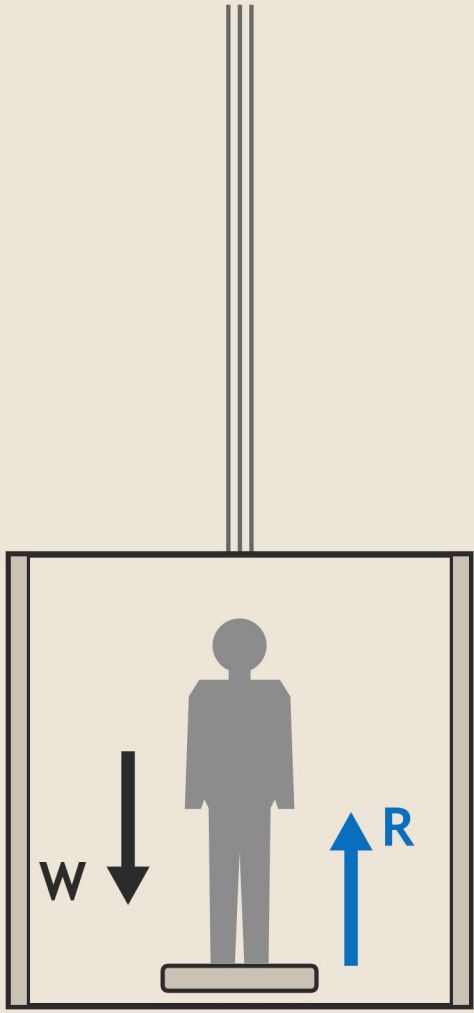
$$a = 0$$

$$R = 588 \text{ N}$$

$$W = 588 \text{ N}$$

- 1 At rest: $R = W$.
- 2 Speeding up: $R > W$.
- 3 Steady speed: $R = W$.
- 4 Slowing down: $R < W$.
- 5 Stopped: $R = W$.

We Do: A Lift Going Down



$$R = 588 \text{ N}$$

$$W = 588 \text{ N}$$

$$v = 0$$

$$a = 0$$

1 Speeding up going down: $R < W$.

2 Steady speed: $R = W$.

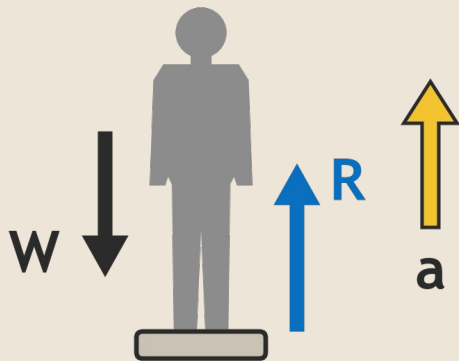
3 Slowing down: $R > W$.

4 Stopped: $R = W$.

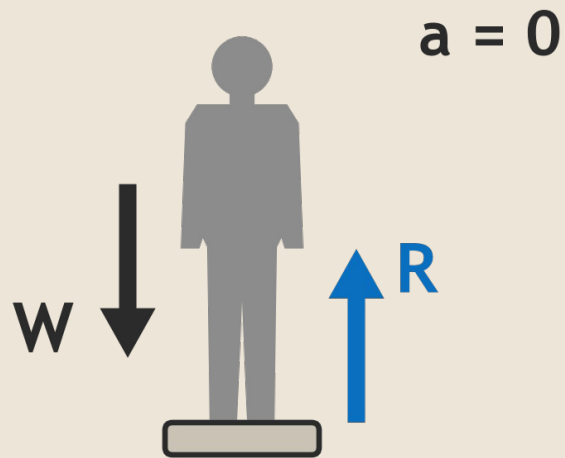
Which Way Is a ?



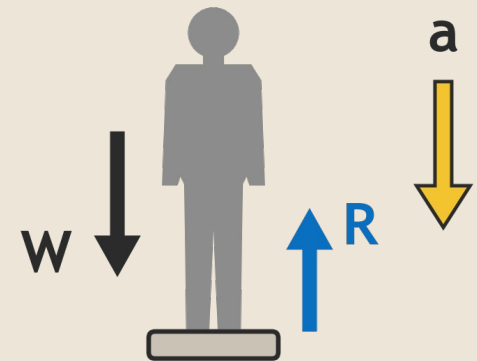
Take up as positive: $R - W = ma$



a up: $R > W$
speeding up going
up, or slowing
going down



$a = 0$: $R = W$
at rest, or steady
speed



a down: $R < W$
speeding up going
down, or slowing
going up

One Ride Up, Two Graphs

Speeding up

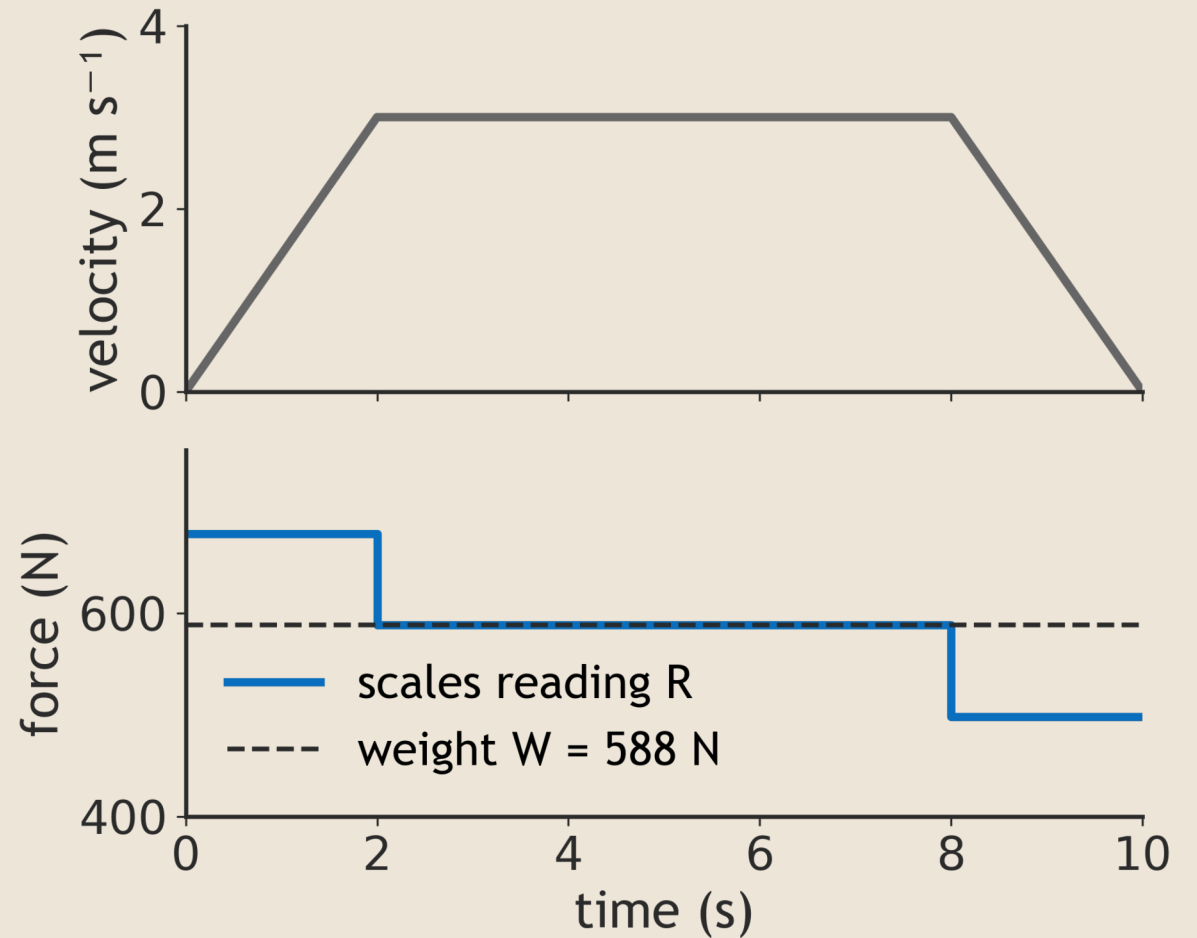
v-t line rises: R above W

Steady

v-t line flat: $R = W$

Slowing down

v-t line falls: R below W



Worked Example: Accelerating Up

A 60 kg pupil is on scales in a lift accelerating up at 1.5 m s^{-2} . Calculate the reading.

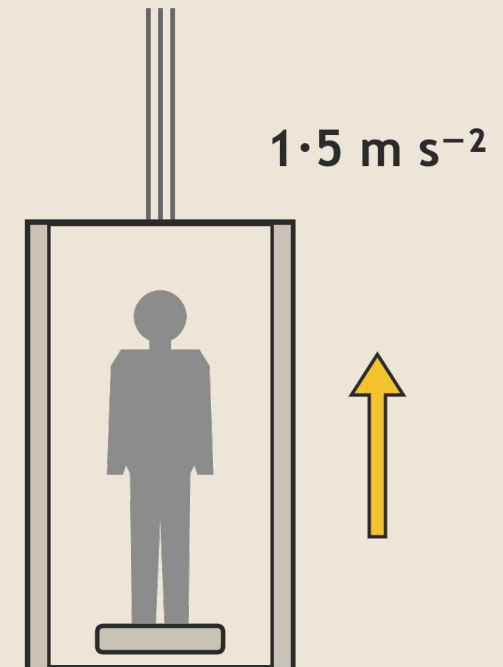
3

$$F = ma \quad (1)$$

$$(W = mg = 60 \times 9.8 = 588 \text{ N})$$

$$R - 588 = 60 \times 1.5 \quad (1)$$

$$R = 678 \text{ N} \quad (1)$$



Worked Example: Accelerating Down

A 45 kg pupil is on scales in a lift accelerating down at 2.0 m s^{-2} . Calculate the reading.

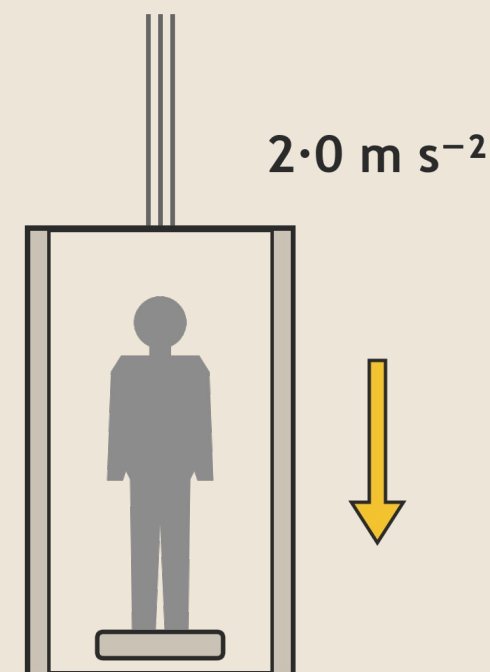
3

$$F = ma \quad (1)$$

$$(W = mg = 45 \times 9.8 = 441 \text{ N})$$

$$R - 441 = 45 \times (-2.0) \quad (1)$$

$$R = 351 \text{ N} \quad (1)$$



Worked Example: Finding a

Scales in a lift read 560 N for a 50 kg pupil. Calculate the lift's acceleration.

3

$$F = ma \quad (1)$$

$$(W = mg = 50 \times 9.8 = 490 \text{ N})$$

$$560 - 490 = 50 \times a \quad (1)$$

$$a = 1.4 \text{ m s}^{-2} \text{ upwards} \quad (1)$$



Your Turn: Scales in a Lift



1

A 70 kg pupil. The lift accelerates upwards at 0.80 m s^{-2} . Calculate the reading.

742 N

2

A 55 kg pupil. The lift slows down at 1.2 m s^{-2} going up. Calculate the reading.

473 N – so a is downwards, $R < W$

3

The scales read 0 N. State how the lift is accelerating.

Downwards at 9.8 m s^{-2} – so falling freely

2015 Section 1 Q4. A person stands on bathroom scales in a lift. The scales show a reading greater than the person's weight. The lift is moving



- A upwards with constant speed
- B downwards with constant speed
- C downwards with increasing speed
- D downwards with decreasing speed
- E upwards with decreasing speed.

Answer D: slowing down going down – so a is upwards and $R > W$

2022 Paper 1 Q5. A student of mass 50.0 kg stands on a newton balance in a lift. The lift accelerates upwards at 1.2 m s^{-2} . The reading on the newton balance is:



A 60 N

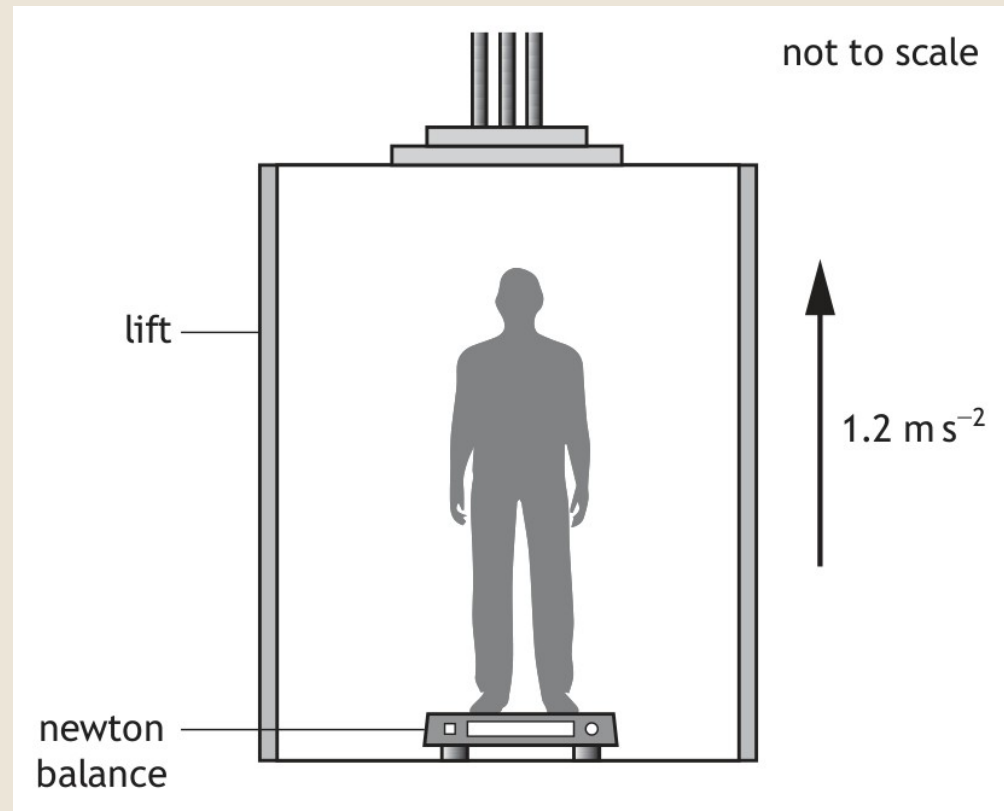
B 430 N

C 490 N

D 550 N

E 590 N

Answer D: $R - 490 = 50.0 \times 1.2$ — so $R = 550\text{ N}$



Spot the mistake. A 60 kg pupil is in a lift accelerating upwards at 2.0 m s^{-2} . What is wrong with this answer?

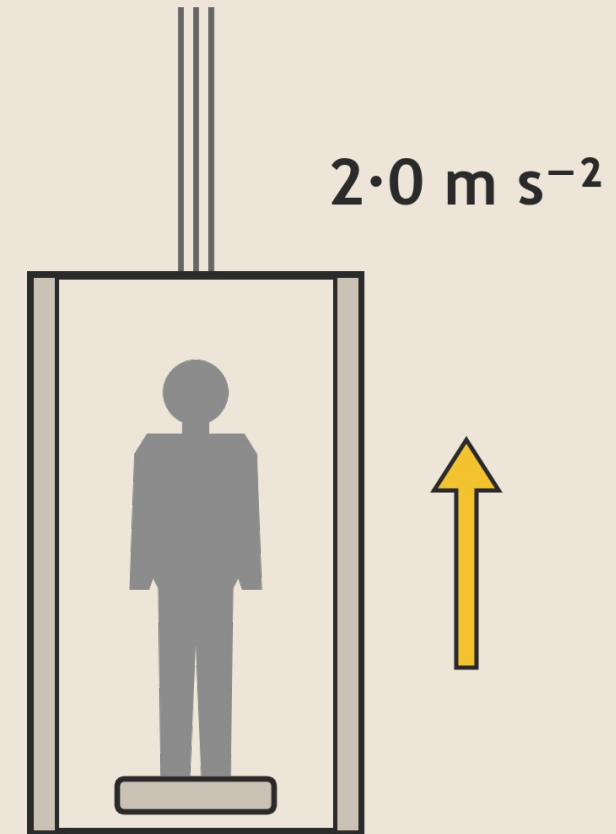


$$R = m(g - a)$$

$$R = 60 \times (9.8 - 2.0)$$

$$R = 468 \text{ N}$$

**a is upwards — so $R > W$: $R = 60 \times 11.8$
 $= 708 \text{ N}$**



Practical: Measure a Lift's Acceleration

You need

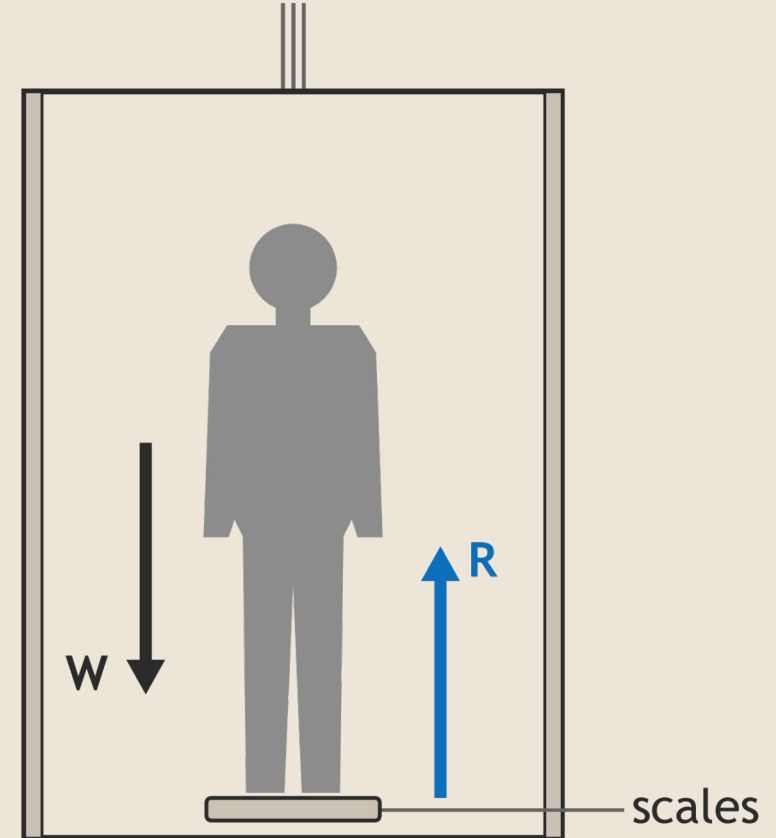
the lift, newton scales, a phone

Measure R

at rest, then on rides up and down

Film it

the reading changes in 1-2 s



Lift Practical



Lift practical sheet

Groups of 3: one rides, one films, one records.

Three rides up, three rides down.

Then check a with phyphox: Acceleration (without g).



SAFETY An adult rides too. Stand still, scales flat.

From Readings to Acceleration

At rest

$$R = W, \text{ so } m = W \div 9.8$$

Each stage

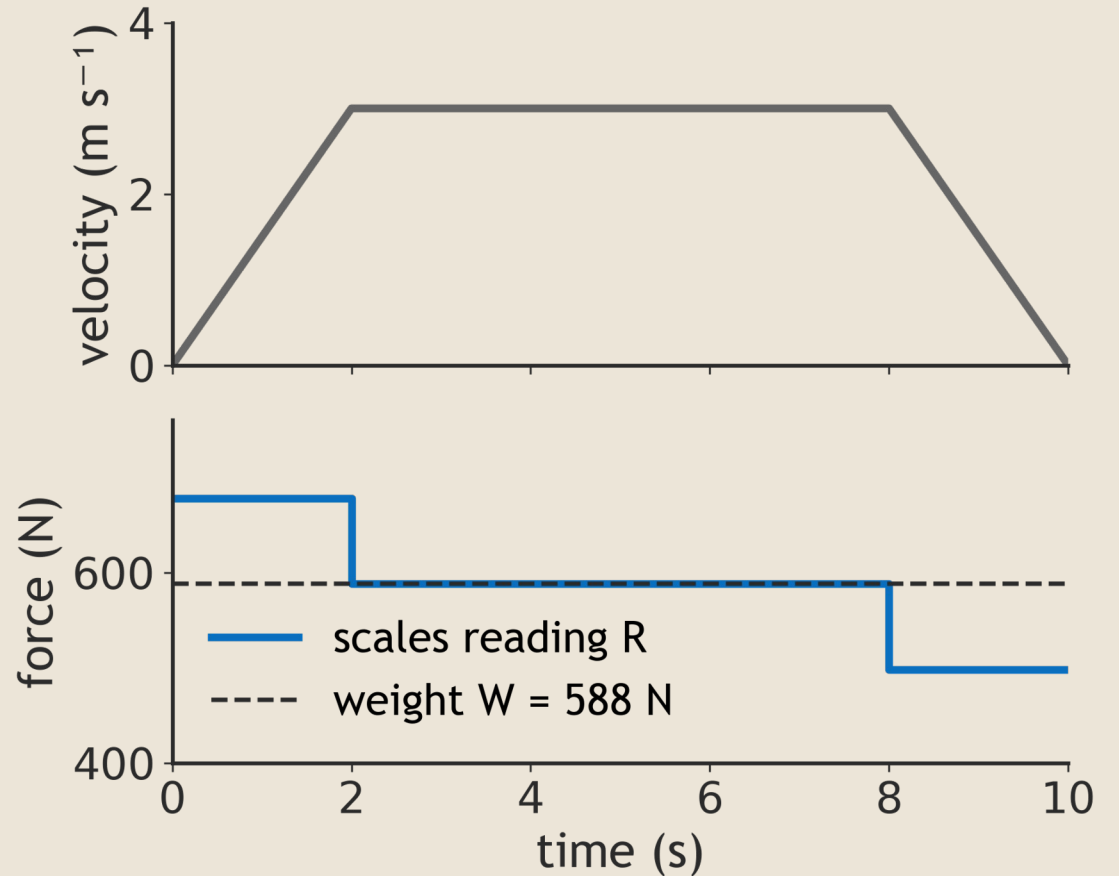
$$a = (R - W) \div m$$

Check

against phyphox

Evaluate

repeat; the scales lag



Plenary: Lifts



- 1 What do the scales read?
- 2 A lift slows down going up. Is R bigger or smaller than W ?
- 3 A 50 kg pupil. $a = 1.0 \text{ m s}^{-2}$ upwards. Calculate the reading.
- 4 Why film the scales display?

R , the upward push on you

Smaller – so a is downwards

540 N

It changes in 1-2 s – so too fast to read live

Look back — by the end of this part...



- 1 I can draw the forces on a person standing on scales in a lift
- 2 I can use $F = ma$ to find a scales reading or a lift's acceleration
- 3 I can describe how to measure a lift's acceleration with newton scales

2010 Higher Physics: scan the code and answer these.



Q2

Q16

Section A · motion, semiconductors

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 4

Friction and terminal velocity

A skydiver falls for a minute, but stops speeding up after about 15 s.

By the end of this part...

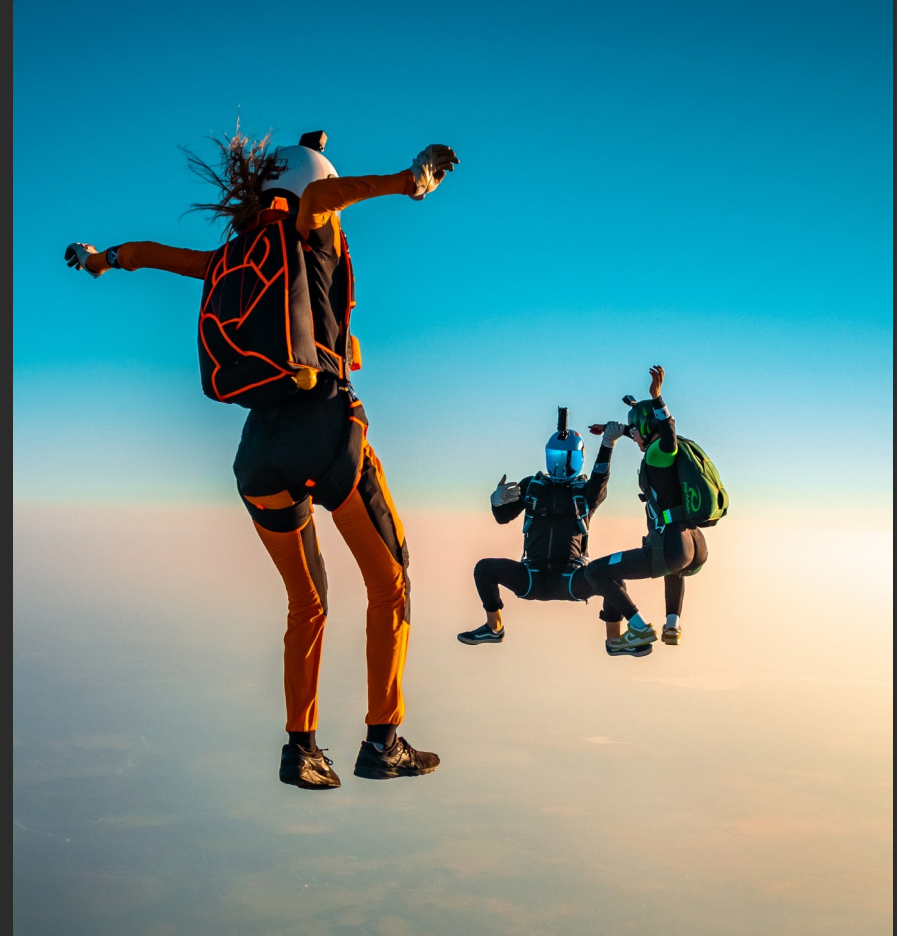


- 1 I can describe the effect of friction on a moving object
- 2 I can explain terminal velocity in terms of forces

Why does a skydiver stop speeding up?



Skydivers in free fall



A formation

Recap from N5: Friction

Friction

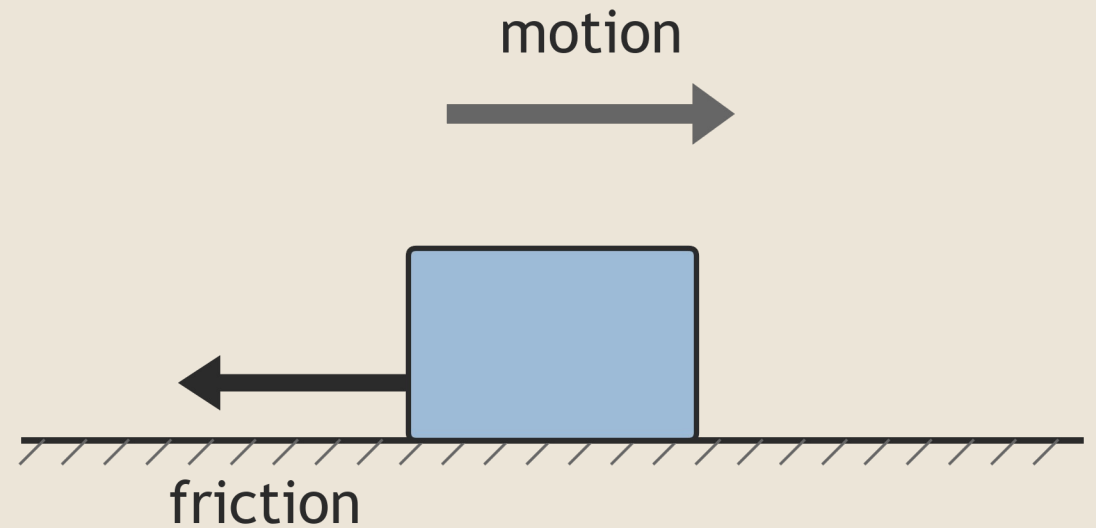
opposes motion between surfaces

Direction

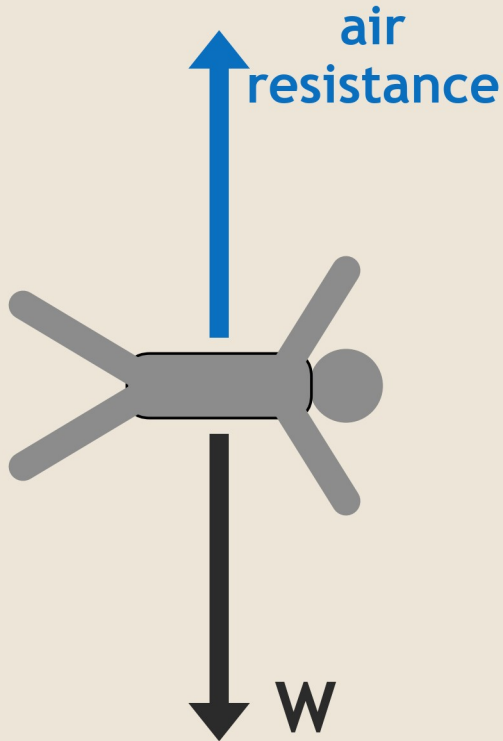
opposite to the motion

Air resistance

friction from the air; grows with speed



Terminal Velocity



$$v = 55 \text{ m s}^{-1}$$

1 Just jumped: only W .
 $a = 9.8 \text{ m s}^{-2}$.

2 Speeding up: air resistance grows.

3 Air resistance nearer W : a smaller.

4 Air resistance = W :
terminal velocity.

Explaining Terminal Velocity



Speed rises

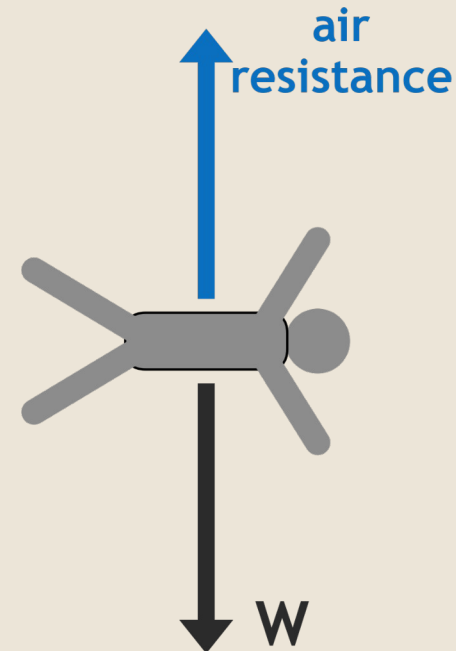
so air resistance rises

Until

air resistance = weight

So

forces balanced: $a = 0$,
constant velocity



$$v = 55 \text{ m s}^{-1}$$

Your Turn: Friction and Terminal Velocity



1

A 70 kg skydiver feels 500 N of air resistance. Calculate their acceleration.

2.7 m s^{-2}

2

A skydiver's weight is 750 N. At terminal velocity, state the air resistance.

750 N – so the forces are balanced

3

Explain why a skydiver's acceleration falls as they speed up.

Air resistance rises – so F falls

2026 Paper 1 Q2. A falling object travels at its terminal velocity because



- A the air resistance is less than the weight of the object
- B the air resistance is balanced by the weight of the object
- C the air resistance is greater than the weight of the object
- D the acceleration of the object is 9.8 m s^{-2}
- E the acceleration of the object is greater than 9.8 m s^{-2} .

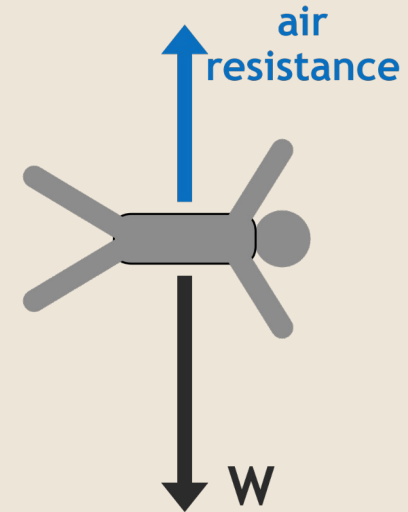
Answer B: balanced forces — so $a = 0$ and v is constant

Spot the mistake. A pupil writes this about a skydiver who has just jumped. What is wrong?



As the skydiver speeds up,
air resistance increases,
so the acceleration increases.

Air resistance opposes W – so the
unbalanced force and a decrease



$$v = 40 \text{ m s}^{-1}$$

Plenary: Friction and Terminal Velocity



1

A skydiver's air resistance is half their weight. State a .

4.9 m s^{-2}

2

At terminal velocity, what is the acceleration?

Zero

3

Why does a skydiver stop speeding up?

Air resistance = W — so
balanced

Look back — by the end of this part...



- 1 I can describe the effect of friction on a moving object
- 2 I can explain terminal velocity in terms of forces

2013 Revised Higher Physics: scan the code and answer these.



Q2

Q4

Q20

Section A · motion, forces, p-n

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 5

Falling with air resistance

One graph tells the whole story of a skydive.

By the end of this part...



- 1 I can interpret a velocity-time graph for a falling object
- 2 I can link each part of the graph to the forces

A Skydiver's v-t Graph



P to Q

air resistance grows

Q to R

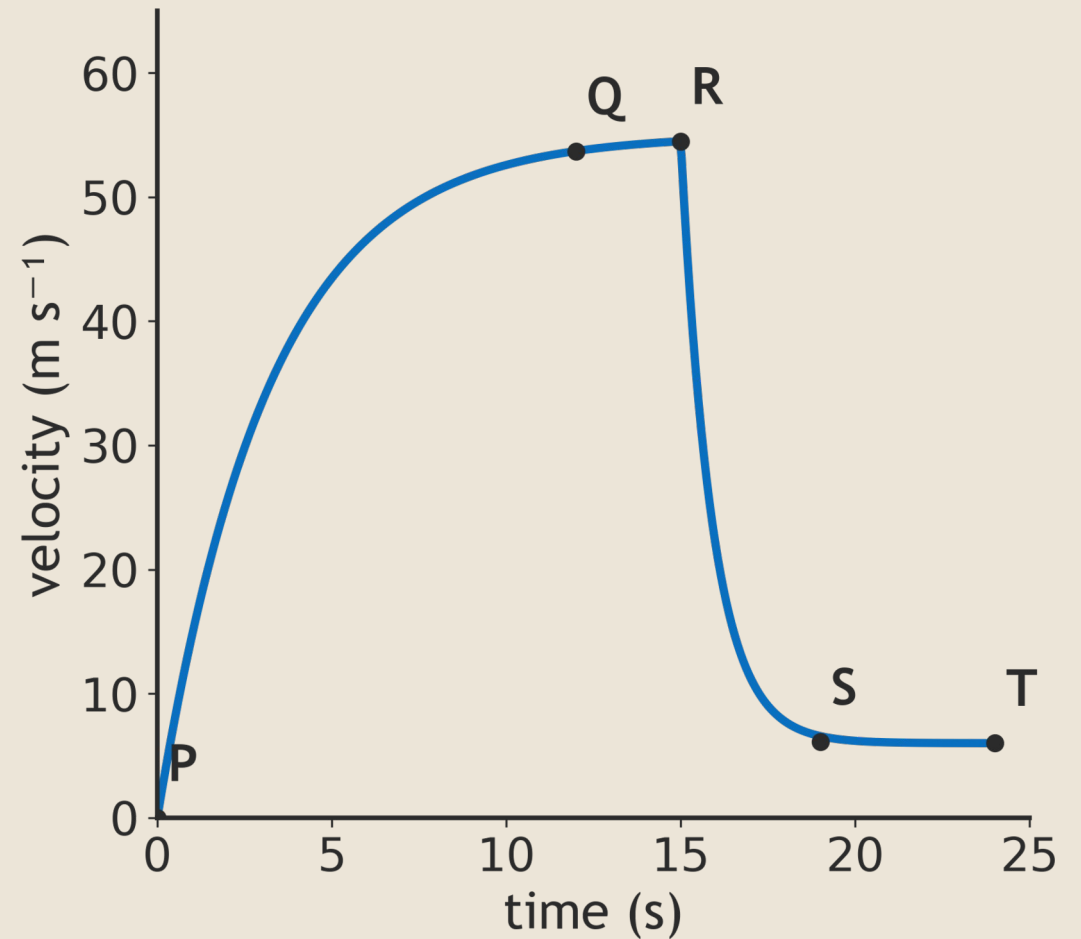
terminal velocity

R to S

air resistance > W: slows

S to T

lower terminal velocity



The Gradient Is the Acceleration

Steep

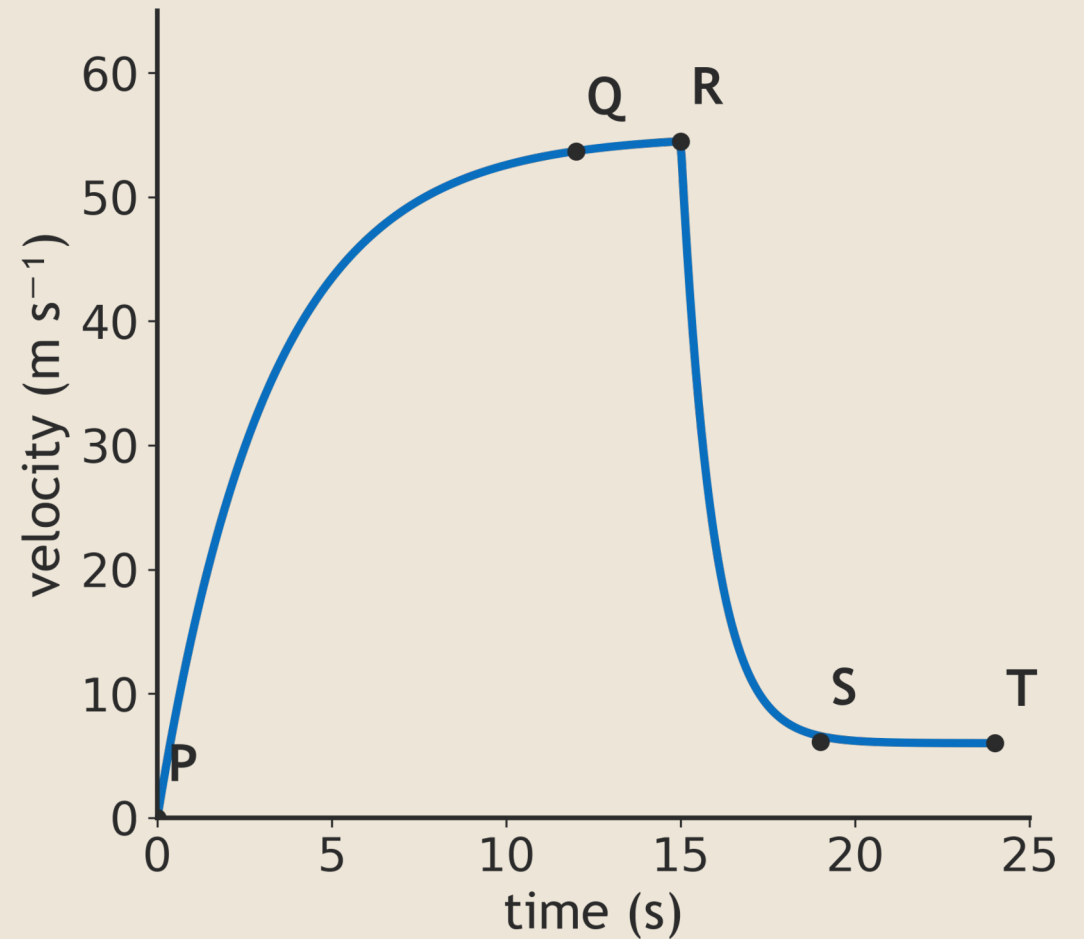
large a : W much bigger than air resistance

Gradient falls

a falls as air resistance grows

Flat

$a = 0$: forces balanced



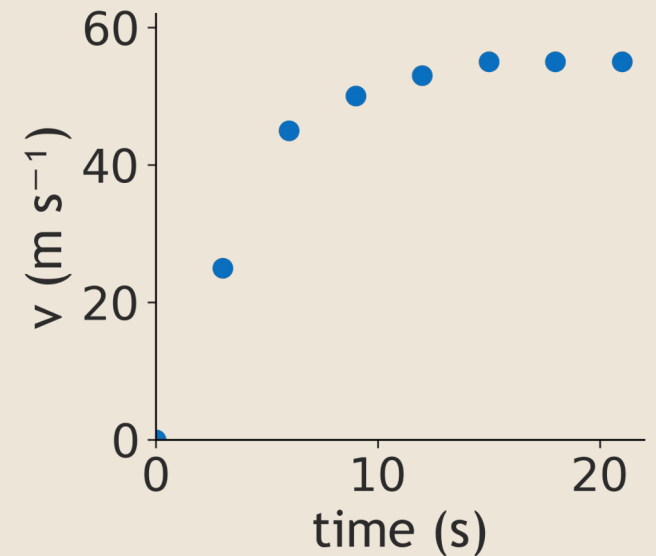
Worked Example: A Skydiver's Data

A skydiver's speed is recorded every 3 s (graph). Explain the forces from 15 s to 21 s.

2

v is constant (55 m s^{-1}), so $a = 0$ (1)

air resistance = weight, so balanced (1)



Your Turn: Skydiver Graphs



- 1 Sketch the v - t graph of an object falling from rest to terminal velocity.
- 2 State what the gradient of a v - t graph shows.
- 3 A parachute opens. Compare air resistance with W .

A curve that levels off

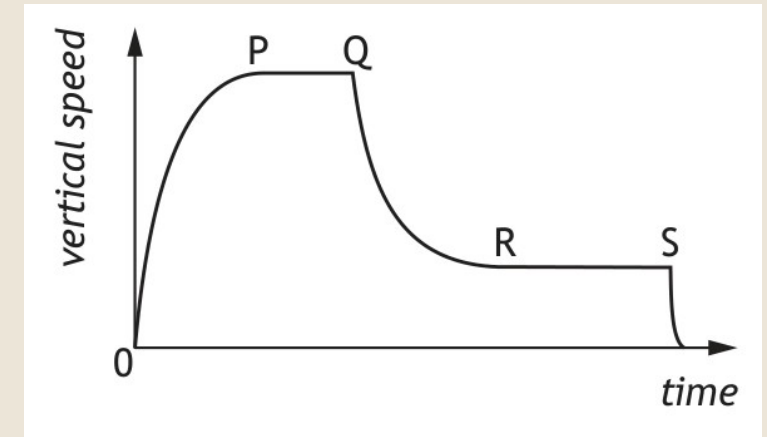
The acceleration

Bigger – so the skydiver slows

2017 Section 1 Q3. A skydiver's vertical speed against time. Which statements are correct?



- I a is greatest from P to Q.
- II Q to R: air resistance $<$ weight.
- III R to S: the forces are balanced.



A I only B II only **C III only** D I, II only E I, II, III

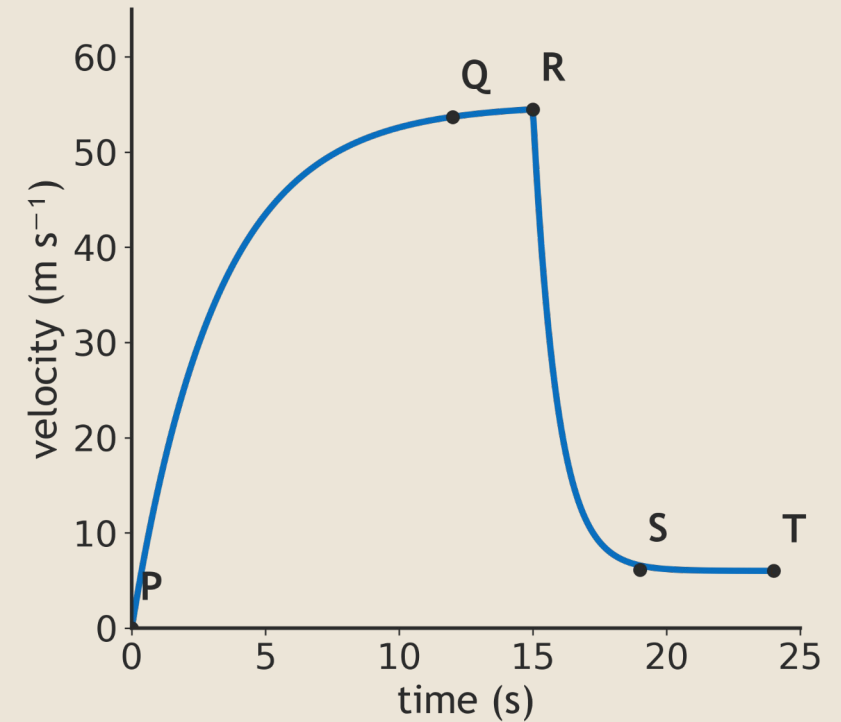
Answer C: P to Q is flat ($a = 0$) and Q to R slows (air resistance $>$ W)
– so only III

Spot the mistake. A pupil matches the a-t graph to a skydiver's v-t graph. What is wrong?



The v-t graph rises, then levels off,
so the a-t graph also rises,
then levels off.

a is the gradient — so a starts at 9.8 m s^{-2} and falls to 0



Plenary: Falling with Air Resistance



1

On a v-t graph, what does a flat line mean?

Terminal velocity — $a = 0$

2

When the parachute opens, which way is the unbalanced force?

Up

3

Is the second terminal velocity higher or lower?

Lower

Look back — by the end of this part...



- 1 I can interpret a velocity-time graph for a falling object
- 2 I can link each part of the graph to the forces

2014 Revised Higher Physics: scan the code and answer these.



Q2

Q17

Q20

Section A · motion, p.d., uncertainty

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 6

Resolving vectors

Split one force into two that are easier to use.

By the end of this part...



- 1 I can resolve a vector into two perpendicular components

Resolving a Force



Horizontal

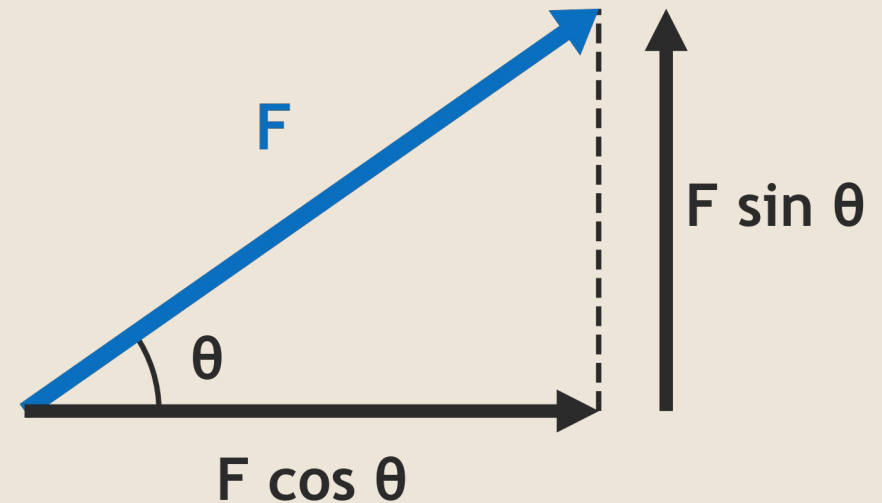
$$F \cos \theta$$

Vertical

$$F \sin \theta$$

θ

measured from the
horizontal



Worked Example: Pulling at an Angle

A 15 N force at 60° above the horizontal pulls a 2.5 kg box along a smooth floor. Calculate its acceleration.

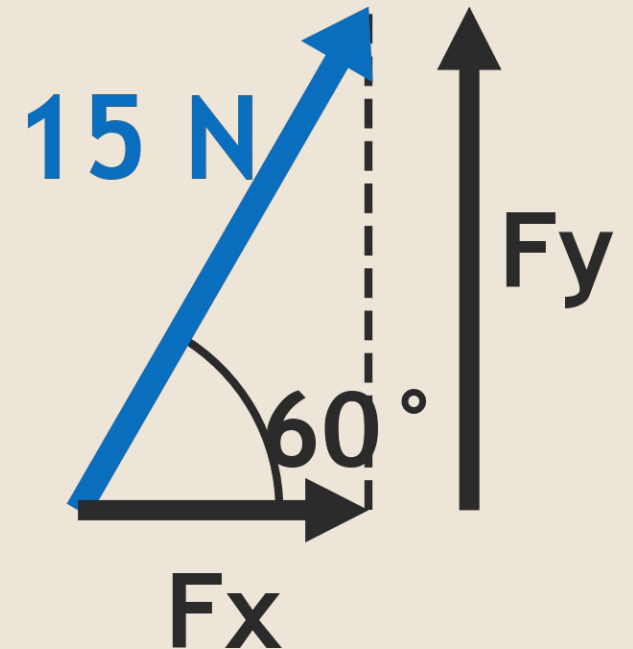
3

$$F = ma \quad (1)$$

$$(\text{component} = 15 \cos 60^\circ = 7.5 \text{ N})$$

$$7.5 = 2.5 \times a \quad (1)$$

$$a = 3.0 \text{ m s}^{-2} \quad (1)$$



Worked Example: A Sledge with Friction

A rope pulls a 15 kg sledge with 80 N at 35° to the horizontal. Friction is 20 N. Calculate its acceleration.

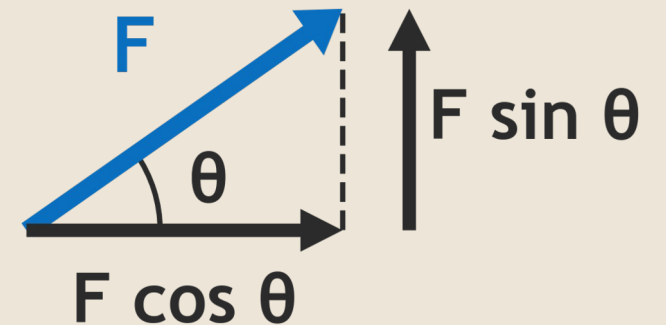
3

$$F = ma \quad (1)$$

$$(\text{component} = 80 \cos 35^\circ = 65.5 \text{ N})$$

$$65.5 - 20 = 15 \times a \quad (1)$$

$$a = 3.0 \text{ m s}^{-2} \quad (1)$$



Your Turn: Resolving



1

A 20 N force at 30° above the horizontal pulls a 4.0 kg block on a smooth floor. Calculate its acceleration.

4.3 m s^{-2}

2

A 12 N force acts at 25° above the horizontal. Calculate both components.

11 N and 5.1 N

3

θ is from the horizontal. Which function gives the vertical part?

$\sin \theta$

2024 Paper 1 Q3. A box hangs from a rope. A horizontal force F holds it stationary as shown. The weight of the box is 4.9 N. The tension T in the rope is



A 1.7 N

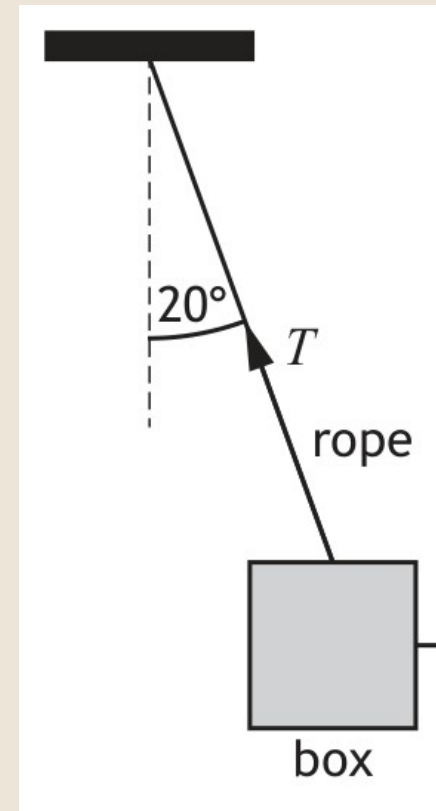
B 4.6 N

C 4.9 N

D 5.2 N

E 14 N

Answer D: $T \cos 20^\circ = 4.9 \text{ N}$
— so $T = 5.2 \text{ N}$



Spot the mistake. A 6.0 N weight hangs from a rope at 30° to the vertical, held by a horizontal force. What is wrong?

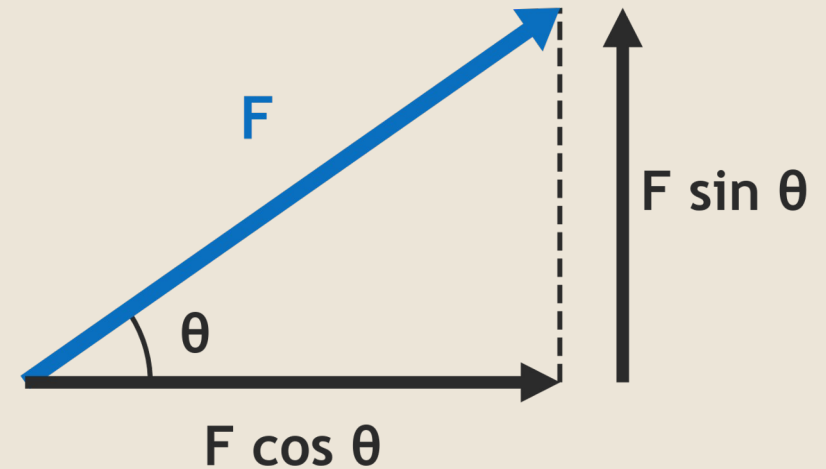


$$T = W \cos 30^\circ$$

$$T = 6.0 \times \cos 30^\circ$$

$$T = 5.2 \text{ N}$$

T's vertical part balances W: $T \cos 30^\circ = 6.0$ — so $T = 6.9 \text{ N}$



Plenary: Resolving



1

Resolve 10 N at 45° above the horizontal.

7.1 N and 7.1 N

2

Which component uses $\cos, \theta$ from the horizontal?

The horizontal one

3

Name the topic in ODU 1 that resolved a velocity.

Projectiles

Look back — by the end of this part...



- 1 I can resolve a vector into two perpendicular components

2009 Higher Physics: scan the code and answer these.



Q2

Q9

Q18

Section A · motion, electricity

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 7

Forces on a slope

On a slope, only part of the weight pulls you down.

By the end of this part...



- 1 I can resolve weight on a slope, parallel and normal to the surface
- 2 I can solve problems on a slope, including friction

Weight on a Slope



Down the slope

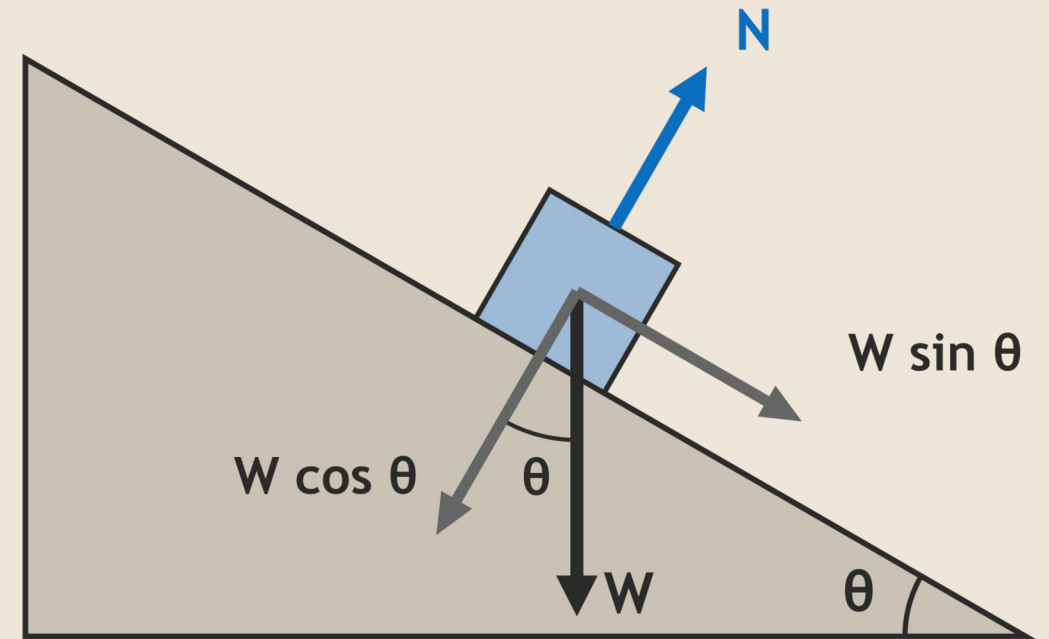
$$W \sin \theta$$

Into the slope

$W \cos \theta$, balanced by N

θ

the slope angle, and between W and the normal



Worked Example: Pulled Up a Slope

A 50 kg crate is pulled up a 20° slope at constant speed.
Friction is 60 N. Calculate the pulling force.

4

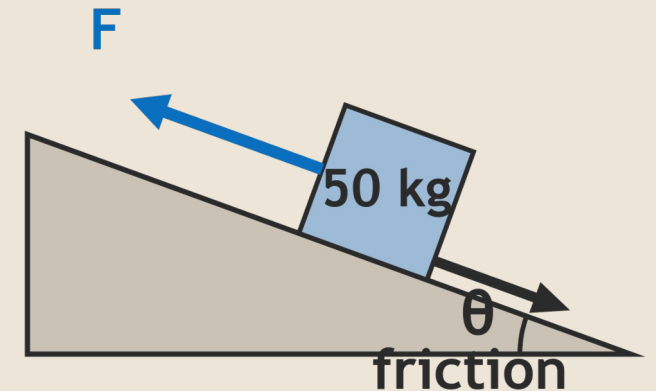
$$F = mg \sin \theta \quad (1)$$

$$F = 50 \times 9.8 \times \sin 20^\circ \quad (1)$$

$$F = 168 \text{ N} \quad (1)$$

(constant speed, so forces balanced)

$$\text{pull} = 168 + 60 = 230 \text{ N} \quad (1)$$



Worked Example: Sliding Down

A 4.0 kg block slides down a 30° slope. Friction is 5.0 N.
Calculate its acceleration.

5

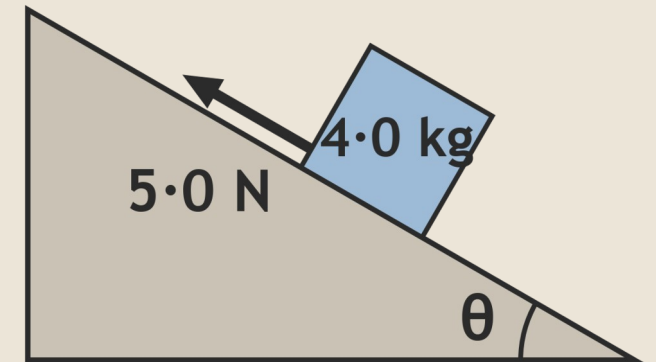
$$F = mg \sin \theta \quad (1)$$

$$F = 4.0 \times 9.8 \times \sin 30^\circ = 19.6 \text{ N} \quad (1)$$

$$F = ma \quad (1)$$

$$19.6 - 5.0 = 4.0 \times a \quad (1)$$

$$a = 3.7 \text{ m s}^{-2} \quad (1)$$



Watch Out: Friction on a Slope

Moving up

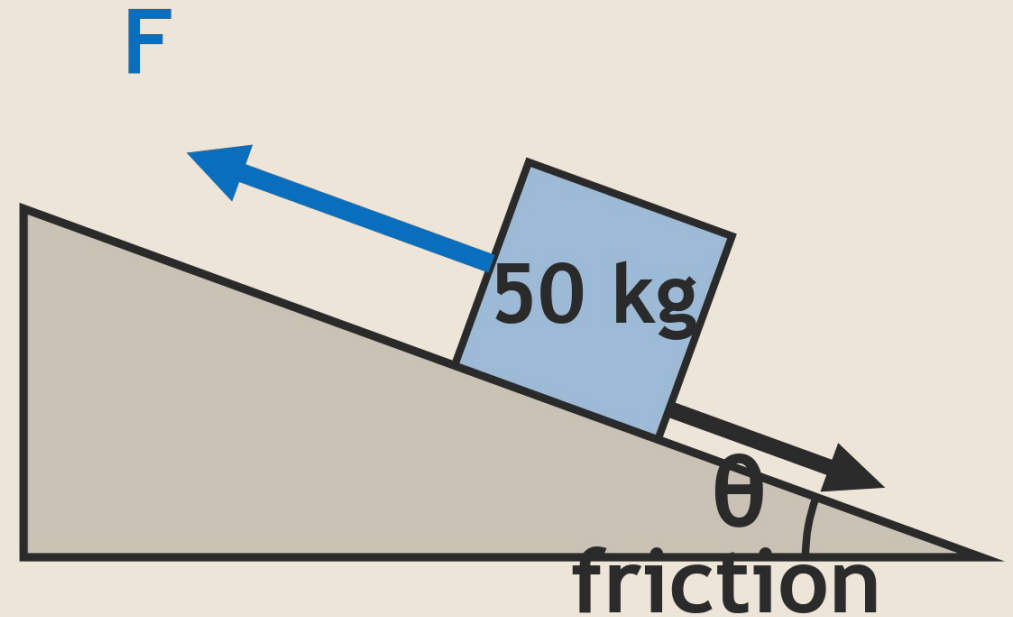
friction acts down: add it

Moving down

friction acts up: subtract it

Always

friction opposes the motion



Your Turn: Slopes



1

A 3.0 kg box slides down a smooth 25° slope. Calculate its acceleration.

$$4.1 \text{ m s}^{-2}$$

2

State which component of the weight N balances.

$$W \cos \theta$$

3

A trolley is pushed up a slope at constant speed. $W \sin \theta = 50 \text{ N}$, friction 20 N. Calculate the push.

70 N – so add the friction

2026 Paper 1 Q3. A cyclist and bicycle (70.0 kg) freewheel down a 15° slope. The component of their combined weight down the slope is



A 18 N

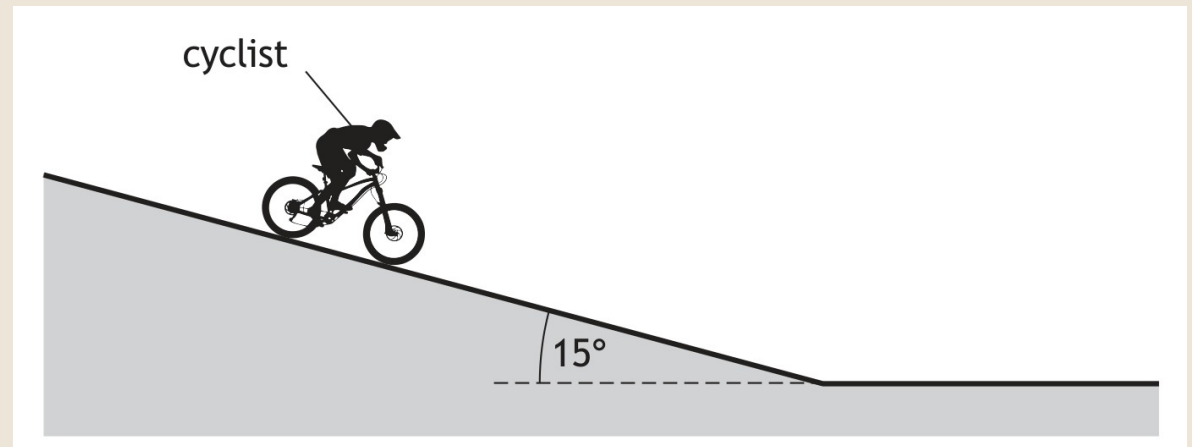
B 68 N

C 180 N

D 660 N

E 690 N

Answer C: $70.0 \times 9.8 \times \sin 15^\circ$ — so 180 N



2022 Paper 1 Q3. A 3.0 kg package moves down a ramp on the Moon ($g = 1.6 \text{ N kg}^{-1}$). Its weight's component parallel to the ramp is:



A 0.89 N

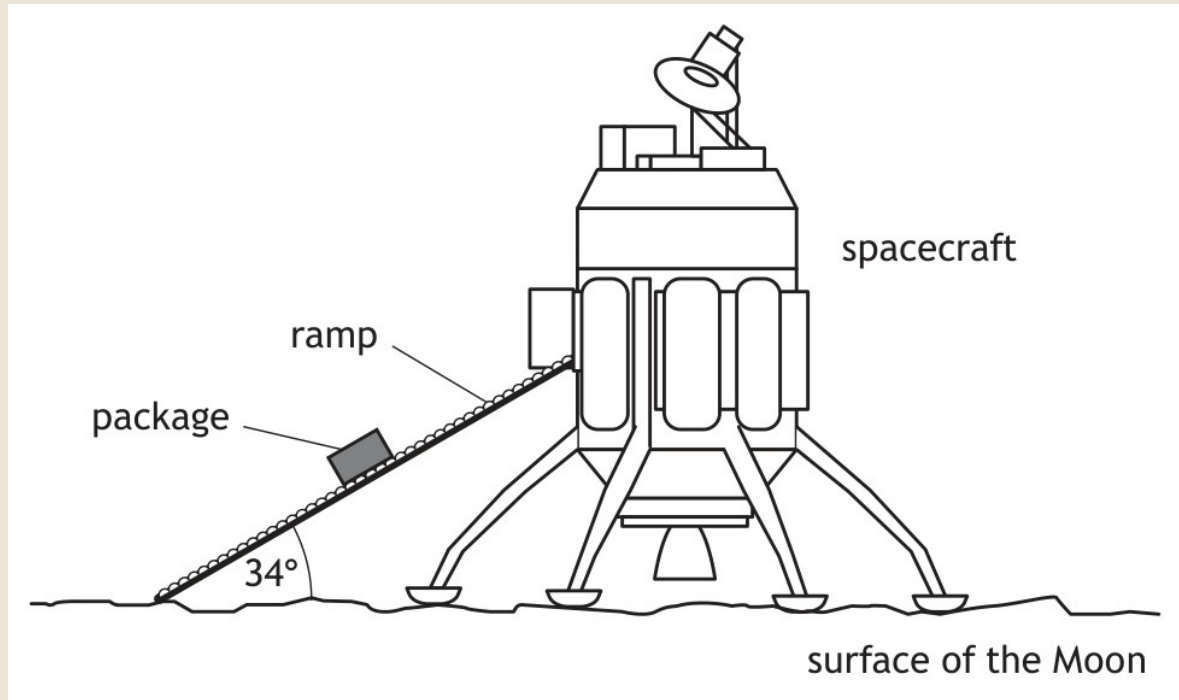
B 2.7 N

C 4.0 N

D 4.8 N

E 16 N

Answer B: $3.0 \times 1.6 \times \sin 34^\circ$ — so 2.7 N



Spot the mistake. A 5.0 kg box is pushed up a smooth 30° slope at constant speed. What is wrong?

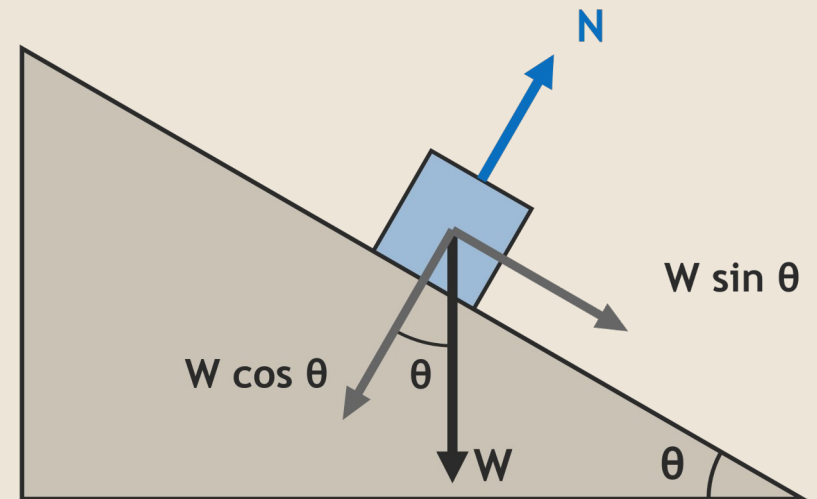


$$F = W$$

$$F = 5.0 \times 9.8$$

$$F = 49 \text{ N}$$

Only $W \sin \theta$ acts down the slope – so F
 $= 49 \times \sin 30^\circ = 25 \text{ N}$



Plenary: Forces on a Slope



1

Which component of weight pulls a box down a slope?

$W \sin \theta$

2

A box slides down. Which way does friction act?

Up the slope

3

Pushed up at constant speed: push = ?

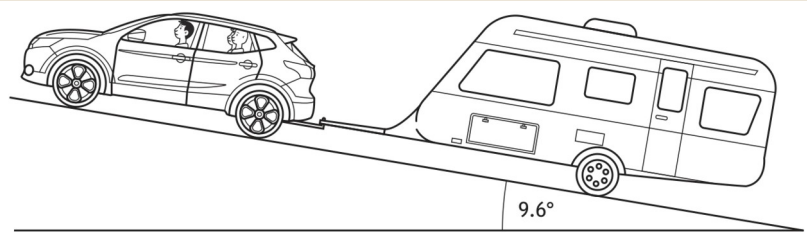
$W \sin \theta + \text{friction}$

Look back — by the end of this part...

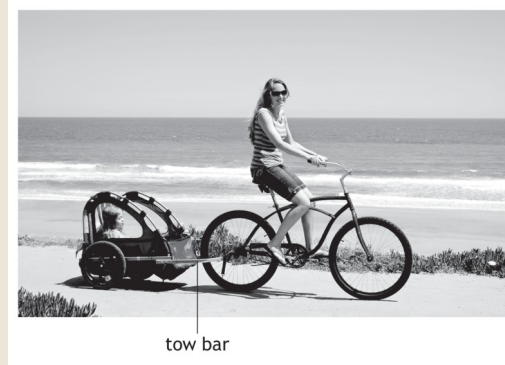


- 1 I can resolve weight on a slope, parallel and normal to the surface
- 2 I can solve problems on a slope, including friction

Forces in the Exam



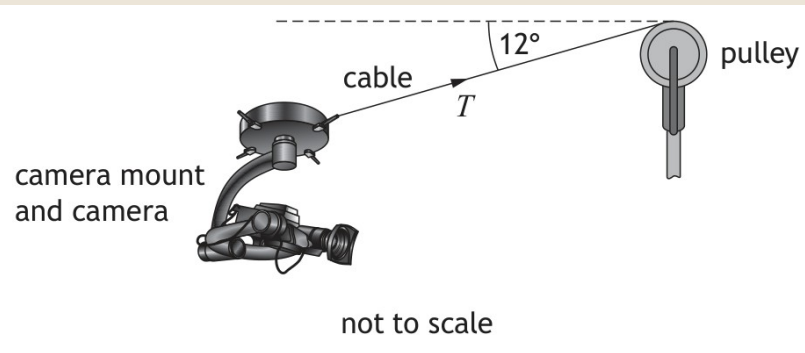
2024 Paper 2 Q2



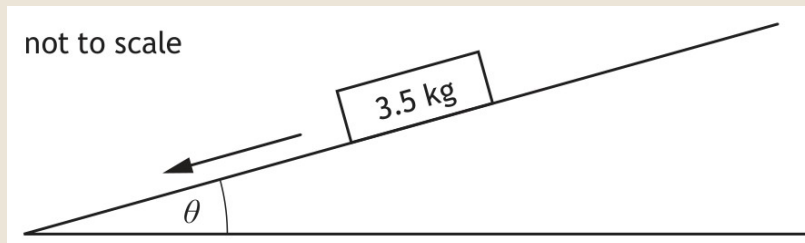
2023 Paper 2 Q2



Past papers



2025 Paper 2 Q1(b)



2025 Paper 1 Q4

Kept for your
test: 2018 P1
Q4, 2016 Q3,
2017 Q2.

Brain Dump



In your jotter

Write everything you remember about forces.
On your own, 2 minutes. No notes.
Then check it against the plenaries.

Extension

EXTRA PRACTICE



1

A car's engine force is constant.
Explain why its acceleration falls as it speeds up.

Air resistance rises –
so F falls

2

Explain why the normal force on a slope is less than the weight.

N balances $W \cos \theta$ –
so $N < W$

LOOKING AHEAD

Next: Energy and Power

Every force that moves something does work.

Course specification — 1 of 3

- Use of vector addition and appropriate relationships to solve problems involving balanced and unbalanced forces, mass, acceleration, and gravitational field strength.
- Knowledge of the effects of friction on a moving object (no reference to static and dynamic friction).
- Explanation, in terms of forces, of an object moving with terminal velocity.

Course specification – 2 of 3

- Interpretation of velocity-time graphs for a falling object when air resistance is taken into account.
- Use of Newton's first and second laws to explain the motion of an object.
- Use of free body diagrams and appropriate relationships to solve problems involving friction and tension.

Course specification — 3 of 3

- Resolution of a vector into two perpendicular components.
- Resolution of the weight of an object on a slope into component forces parallel and normal to the surface of the slope.

Energy and power, the last bullet, is deck 2b.

Credits (1 of 3)

Starter: 2012 Revised Higher Physics, © Qualifications Scotland (SQA); 2015 Higher Physics, © Qualifications Scotland (SQA); 2014 Higher Physics, © Qualifications Scotland (SQA); 2010 Higher Physics, © Qualifications Scotland (SQA); 2013 Revised Higher Physics, © Qualifications Scotland (SQA); 2014 Revised Higher Physics, © Qualifications Scotland (SQA); 2009 Higher Physics, © Qualifications Scotland (SQA)

About Isaac Newton: Portrait: Godfrey Kneller, 1689, public domain, via Wikimedia Commons

2025 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2025, Paper 1, Question 3.

2015 Section 1 Q5: © Qualifications Scotland (SQA). Higher Physics 2015, Section 1, Question 5.

The big question (A lift): © David Anstiss, CC BY-SA 2.0, via Geograph

The big question (A drop tower ride): Photo: Stefan Scheer, CC BY-SA 4.0, via Wikimedia Commons

Credits (2 of 3)

2015 Section 1 Q4: © Qualifications Scotland (SQA). Higher Physics 2015, Section 1, Question 4.

2022 Paper 1 Q5: © Qualifications Scotland (SQA). Higher Physics 2022, Paper 1, Question 5 (figure and text).

The big question (Skydivers in free fall):
Photo: Rstpch, CC BY-SA 4.0, via Wikimedia Commons

The big question (A formation): Photo:
Rstpch, CC BY-SA 4.0, via Wikimedia Commons

2026 Paper 1 Q2: © Qualifications Scotland (SQA). Higher Physics 2026, Paper 1, Question 2.

2017 Section 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2017, Section 1, Question 3.

2024 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2024, Paper 1, Question 3.

2026 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2026, Paper 1, Question 3.

Credits (3 of 3)

2022 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2022, Paper 1, Question 3.

Forces in the Exam: © Qualifications Scotland (SQA). Higher Physics 2023 Paper 2 Q2, 2024 Paper 2 Q2, 2025 Paper 1 Q4 and Paper 2 Q1.

Course specification — 1 of 3: © Qualifications Scotland (SQA). Higher Physics Course Specification v3.0, p.7

Course specification — 2 of 3: © Qualifications Scotland (SQA). Higher Physics Course Specification v3.0, p.7

Course specification — 3 of 3: © Qualifications Scotland (SQA). Higher Physics Course Specification v3.0, p.7